$= N_0 e^{-\lambda t}$, which indicates the number of the atoms not yet decayed during the time *t*. The energy *E* can be computed, from the range, by means of a law also discovered by Geiger and Nuttall.

Figure 20 shows the relationship between logλ and *E* for all alpha emitters, in the form of a diagram based on measurements. We see here three adjacent curves, one for each radioactive series—the uranium series, the thorium series and the actinium series. Although these lines are not perfectly straight, as the Geiger-Nuttall relationship would require, they are at least not too strongly curved. The very fact that we get three different, approximately parallel curves, shows that although the constant *B* is the same for all the three radioactive series, the values of the constant *A* differ slightly. At the very bottom we find uranium, with the smallest decay probability, and at the very top is radium C' with the highest such probability.

The point now is to explain this regularity, and above all, the fact—very astonishing at first glance—that such a small change in the energy, *E*, produces such an enormous change in the decay probability, λ. In this entire range, the energy varies only between 6×10^{-6} and 13×10^{-6} ergs—a ratio of 1 to 2. On the other hand, the decay probability varies between the orders of magnitude of 10^{-18} and 10^{6} per second—a ratio of 1 to 10^{24}.

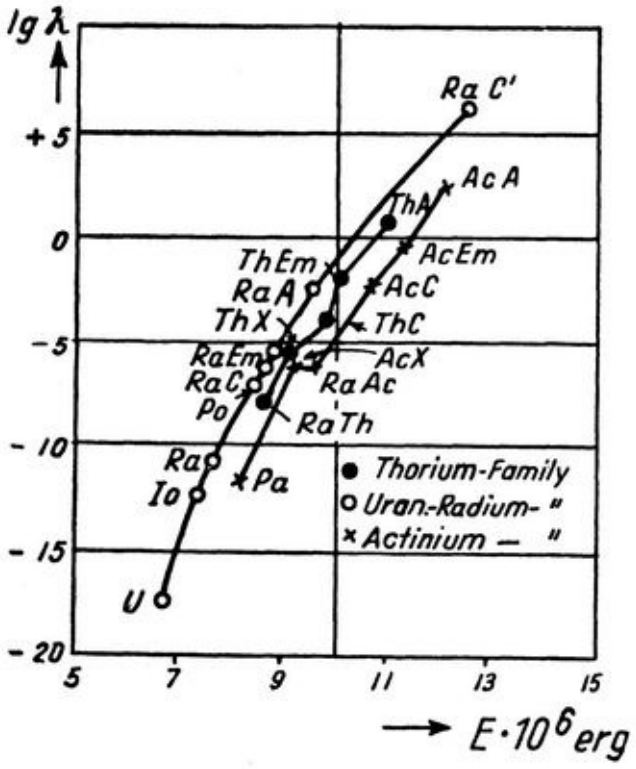


Figure 20.—Illustrating the Geiger-Nuttall law.

This fact was explained by Gamow, Condon and Gurney, in 1928. In order